

Md. Forhan Shahriar Fahim

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RESEARCH INTERESTS

AI Safety, AI Interpretability, Large Language Models, Computer Vision, Medical Imaging

EDUCATION

University of Rajshahi

Jan 2019 – Dec 2024

B.Sc. in Computer Science & Engineering (CGPA: 3.66, Last 2 Years Avg: 3.80)

Rajshahi, Bangladesh

- **Thesis:** Force Invariant EMG Pattern Recognition Using Deep Learning (Supervisor: Dr. Shamim Ahmad)
- **Relevant Coursework:** Data Structures (C++), Algorithms, OOP (Java), Database Systems, Operating Systems, AI, Digital Image Processing, Software Engineering, System Analysis.

EXPERIENCE

Pundra University of Science & Technology

Mar 2025 – Present

Lecturer

Bogura, Bangladesh

- Teach undergraduate courses in Design and Analysis of Algorithms and Machine Learning; have also taught Web Engineering, Structured Programming, and Simulation & Modeling.
- Develop course materials and assessments aligned with Outcome-Based Education (OBE).
- Supervise undergraduate research theses in AI and Machine Learning, guiding students through problem formulation, experimental design, model evaluation, and manuscript preparation.

AMIR Lab (Advanced Machine Intelligence Research Lab)

May 2025 – Dec 2025

Research Intern

- Screened and analysed studies for a PRISMA-guided review of self-supervised learning in cancer imaging.
- Drafted sections and figures for the review manuscript.

PUBLICATIONS

- [1] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Nafiu Rahman, and Adrita Alam, “VulPatchNet: A Dual-Encoder Cross-Attention Framework for Multi-Language Vulnerability Patch Generation,” in *5th IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)* (Accepted).
- [2] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, and Md. Tanjid Sheikh, “Explainable Multi-Task Transformer-Based Framework for Cross-Language Source Code Vulnerability Detection with Auxiliary CWE and Language Classification,” *Journal of Cyber Security Technology*, vol. 10, no. 1, Art. no. 2700412, 2026. doi:10.1080/23742917.2026.2700412 [🔗](#).
- [3] Md. Tanjid Sheikh, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, Sumaya Akter Mahmuda, and S. M. Imran Reza, “Systematic Hybrid Analysis of Android Applications: CWE-Categorized Vulnerabilities and Practical Mitigation Strategies,” in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, pp. 1–6, 2026. doi:10.1109/QPAIN69676.2026.11546396 [🔗](#).
- [4] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, and Md. Tanjid Sheikh, “An Empirical Analysis of Transformer-Based Models with LIME Explainability for JavaScript Vulnerability Detection,” in *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, pp. 1–6, 2026. doi:10.1109/ICECTE69292.2026.11429428 [🔗](#).

RESEARCH EXPERIENCE

Neuron-Level Interpretability in Vision Transformers | Ongoing Research

Mar 2026 – Present

- Investigate neuron activation patterns in vision transformers to characterize learned concepts and feature selectivity.
- Apply neuron visualization and quantitative analysis methods to examine interpretability and explore its relevance to downstream tasks.

Systematic Review: SSL in Cancer Imaging | *Lead Researcher*

Aug 2025 – Sep 2026

- Reviewed Self-Supervised Learning (SSL) methods across MRI, CT, and histopathology using a PRISMA-guided approach to study label-efficient medical image analysis.
- Synthesized approaches, datasets, evaluation strategies, and research gaps to inform a manuscript for peer-reviewed publication.

Force-Invariant EMG Pattern Recognition Using Deep Learning | *Thesis*

Apr 2023 – Sep 2024

- Developed a hybrid CNN-LSTM framework for force-invariant sEMG pattern recognition, benchmarking its performance against ResNet-50 and VGG-16 under varying contraction forces.
- Evaluated classification performance across varying contraction force levels, achieving 80% accuracy.

TECHNICAL PROJECTS

CSE Academic Operations Hub [🔗](#) | *Next.js, TypeScript, PostgreSQL*

Aug 2026 – Sep 2026

- Built a portal for scheduling, attendance, and faculty workload across HSC and Diploma cohorts.
- Blocked conflicting routines from publication with minute-level teacher, room, and cohort checks.
- Added CSV roster import, attendance summaries, extra-class payment sheets, and versioned exports.

ACADEMIC SUPERVISION

- Undergraduate Research Supervisor (10+ students):** Supervise undergraduate research in medical imaging, including *Brain Tumor Segmentation using Attention Mechanisms* and *Automated Tumor Detection*.
- Capstone Project Supervisor:** Supervise full-stack development projects, including an *Artisanal E-commerce Platform* and a *Web-based Hostel Management System*, with emphasis on software development life cycle (SDLC), system design, implementation, and agile documentation.

PROBLEM SOLVING

Competitive Programming | *Codeforces, LeetCode, AtCoder*

- Solved 1,000+ algorithmic problems across various platforms; achieved **Max Rating 1396 (Pupil)** on Codeforces and **1661** on LeetCode.

HONORS & AWARDS

- Young Enthusiastic Faculty Member Award (2025), Pundra University of Science & Technology:** Recognized for teaching, student mentorship, and departmental service.
- Dean's Award (2023), University of Rajshahi:** Recognized for outstanding undergraduate academic performance.

TECHNICAL SKILLS

- Programming:** Python, C/C++, Java.
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face.
- Software & Databases:** Next.js, Node.js, PostgreSQL, MySQL.
- Tools:** Git, Linux, LaTeX.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- Event Coordinator, PUB CSE Fest (2026):** Coordinated technical events and supported broader festival planning, sports logistics, and media coverage.
- Competitive Programming Mentor:** Mentored students through the PUB and RU programming communities, conducting workshops on algorithms, problem solving, and national programming contest preparation.

REFERENCES

Dr. Shamim Ahmad

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University of Rajshahi

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Undergraduate thesis supervisor

Dr. Md. Ekramul Hamid

Professor, Dept. of CSE,
University of Rajshahi

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